

CIS 5300 Milestone 3

Zhirui Bian, Qianye Ding, Maya Gambhir, Andy Li

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Extension Overview

Reward Model Integration

Our extension integrates a Reward Model (RM) into the inference pipeline to improve story continuation quality through best-of-N reranking. The RM learns an explicit scoring function $r(x, y)$ that approximates GPT-4.1-mini’s rubric-based evaluation: at inference time, we sample $N = 3$ candidate continuations and select the one with the highest predicted reward.

Reward Model Training

We construct 3,000 training samples: 2,000 GPT-4-generated continuations (querying twice per prompt for 1,000 stories) plus 1,000 lower-scoring outputs from the base DeepSeek-R1-Distill-Llama-8B model. Each sample pairs the story beginning (excluding instruction tokens) with a continuation and its GPT-4.1-mini rubric scores across five dimensions: Information Completeness (IC), Factual Accuracy (FA), Relevance (REL), Logical Coherence (LC), and Creativity (CE). The RM uses a regression head on Llama-8B with 4-bit quantization + LoRA (batch_size=1, grad_accum=32, epochs=1), predicting `overall_raw` (mean of five scores) as target.

Modified Scoring Formula

To combat verbose, low-quality outputs that “fake quality with length,” we introduce a quality-gated length bonus:

$$\text{overall_quality} = \text{overall_raw} + \lambda \cdot \text{bonus} - \text{penalty}$$

where `bonus` $\in [0, 1]$ rewards answers between 40–120 words, and the quality gate $\lambda = ((\text{IC} + \text{REL})/10)^2$ ensures only high-quality answers benefit from length. A verbosity penalty applies when length exceeds 120 words with low relevance or coherence.

Note that since the evaluation rubric and scoring formula were updated from our previous extension, we re-evaluated both the simple baseline (loose prompt) and strong baseline (strict prompt) under the new criteria to ensure fair comparison.

Empirical Evaluation

We evaluate on 500 randomly sampled ROCStories test examples using GPT-4.1-mini as judge (see Table 1 and Figure 1). We compare: (1) loose prompt baseline, (2) strict CoT prompt, and (3) strict + RM reranking.

Analysis

Adding RM reranking yields significant improvement in Relevance (+0.21) and modest gains in Information Completeness (+0.09), while Factual Accuracy decreases slightly (−0.27). Overall quality remains stable (~ 3.50), demonstrating RM as a robust approach: with three candidates, the lightweight LoRA-based RM performs effective filtering. The primary trade-off is reduced Creativity (−0.37), but this dimension has limited relevance to hallucination suppression—our core objective. Combined with the quality-gated bonus/penalty mechanism, RM reranking enhances robustness at scale.

Table 1: Evaluation results on 500 ROCStories test examples. All scores are on a 1–5 scale.

Method	IC	FA	REL	LC	CE	Raw	Quality
Loose	1.524	4.893	3.264	3.090	2.511	3.056	2.955
Strict	2.177	4.819	3.763	3.439	2.464	3.332	3.522
Strict + RM	2.270	4.549	3.969	3.341	2.091	3.244	3.503

IC: Information Completeness, FA: Factual Accuracy, REL: Relevance, LC: Logical Coherence, CE: Creativity Expression, Raw: Overall Raw, Quality: Overall Quality (with quality-gated length adjustment).

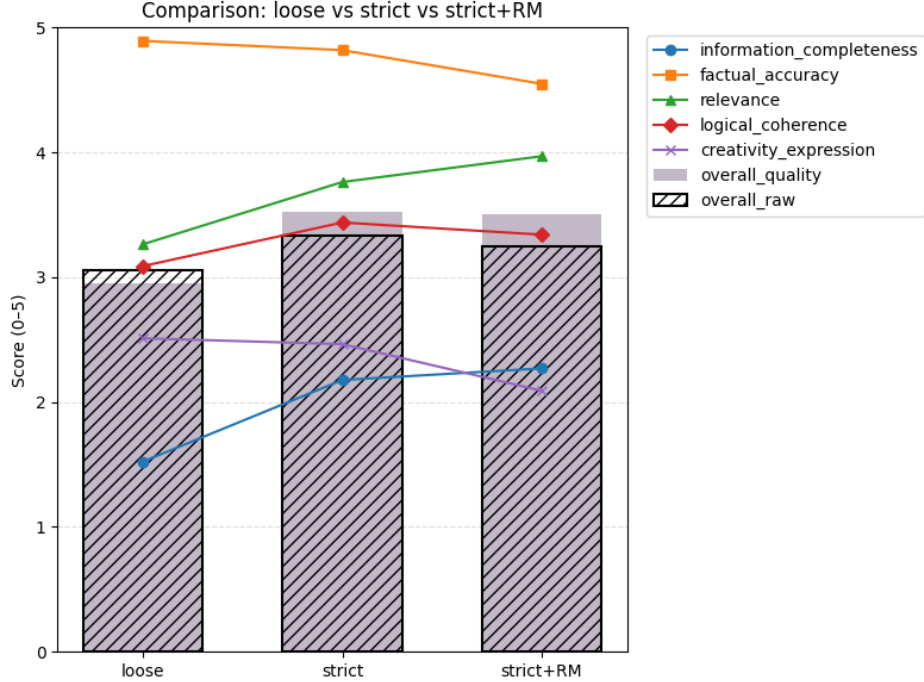


Figure 1: Comparison of evaluation metrics across loose prompting, strict CoT prompting, and strict + RM reranking. Bar heights represent overall scores; lines show individual dimension trends.

Future Work

Direct Preference Optimization

We plan to integrate Direct Preference Optimization (DPO), which directly shapes the policy distribution from pairwise comparisons (x, y^+, y^-) without inference-time sampling overhead. We have constructed 1,000 preference pairs (high-scoring vs. local model outputs) for DPO training. The combined approach leverages complementary strengths: DPO improves one-shot generation quality, while RM exploits sampling through candidate reranking.